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05

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

In 1934 physicist Eugene Wigner posited the existence of a crystal consisting entirely of electrons in a honeycomb-like structure. The so-called Wigner crystal remained largely conjecture, however, until Feng Wang and colleagues announced in 2021 that they had captured an image of one. The researchers trapped electrons between two semiconductors and then cooled the apparatus, causing the electrons to settle into a crystalline structure. By inserting an ultrathin sheet of graphene above the crystal, the researchers obtained an impression—the first visual confirmation of the Wigner crystal.

Which choice best states the main idea of the text?

- A. Researchers have obtained the most definitive evidence to date of the existence of the Wigner crystal.
- B. Researchers have identified an innovative new method for working with unusual crystalline structures.
- C. Graphene is the most important of the components required to capture an image of a Wigner crystal.
- D. It's difficult to acquire an image of a Wigner crystal because of the crystal's honeycomb structure.

In 2016 biological anthropologist Heather F. Smith and her team investigated the evolution of the appendix, an intestinal organ that is present in some mammals, including humans, but is generally thought to have no function. Studying 533 mammal species, the team found that the appendix has emerged independently across multiple lineages in separate instances and, significantly, hasn't disappeared after emerging in specific lineages. Moreover, the team determined that species with the organ tend to have higher concentrations of lymphoid tissue, which supports immune responses, in the cecum, the organ the appendix is attached to. Therefore, the team hypothesized that the appendix likely _____ blank

Which choice most logically completes the text?

- A. was once present in many nonmammal species but has since disappeared from those lineages.
- B. has been preserved in certain mammal species because it benefits their immune systems.
- C. will emerge in a greater number of mammal species because it may serve a necessary function in the immune system.
- D. produced higher concentrations of lymphoid tissue in mammals in the past than it does currently.

Baltimore, Maryland, has installed engineered structures along 71% of its shoreline to protect infrastructure from wave erosion and other hazards, a practice known as shoreline hardening. To evaluate the responses of waterbirds to two types of hardening structures—riprap and bulkheads—Diann Prosser et al. surveyed waterbird communities consisting of the tundra swan, the great blue heron, and 62 other species at different sites in the Chesapeake Bay on the US East Coast. Utilizing the Index of Waterbird Community Integrity (IWCI), on which a high score corresponds to high community integrity, the researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than is riprap.

Which finding, if true, would most directly illustrate the researchers' finding?

- A. The difference in average IWCI scores for waterbird communities at Stony and Old Road, two sites with a higher percentage of shoreline consisting of bulkheads than of riprap, was statistically insignificant.
- B. Waterbird communities at Old Road, a site with a relatively high percentage of shoreline consisting of bulkheads, had lower average IWCI scores than did waterbird communities at Miles, a site with a relatively high percentage of shoreline consisting of riprap.
- C. Waterbird communities at Curtis, a site with a high percentage of shoreline consisting of bulkheads and riprap, had lower average IWCI scores than did waterbird communities at Onancock, a site with a low percentage of shoreline consisting of bulkheads and riprap.
- D. Waterbird communities at Curtis, a site with equal percentages of shoreline consisting of bulkheads and riprap, had higher average IWCI scores than did waterbird communities at Miles, a site with different percentages of shoreline consisting of bulkheads and riprap.

The following text is from Ahmet Hamdi Tanpınar's 1961 novel *The Time Regulation Institute* (translated from the Turkish in 2014 by Maureen Freely and Alexander Dawe). The narrator was once involved with the Time Regulation Institute, a fictional bureaucracy that regulates the time of Turkey's clocks.

I may be the most humble and absurd man in the world and, as my wife says, the most slovenly creature you may ever meet—that is, before the founding of our institute—but I did come to know a truly great man who possessed a natural genius for invention. I spent years at his side. I watched the way he worked. I witnessed how an idea would suddenly catch fire in his mind and take shape, like a tree sprouting shoots and branches, before coming into being.

©2014 by Maureen Freely and Alexander Dawe

Which choice best states the main idea of the text?

- A. A person was fortunate enough to have found love despite his obvious shortcomings.
- B. An exceptionally talented person professes humility to avoid intimidating others.
- C. A person appreciates the extensive time he passed in the company of someone he deeply admired.
- D. A truly brilliant person lets ideas mature in his mind before expressing them aloud.

It's common for jazz musicians and fans to refer to certain songs as having "swing," indicating that the songs provoke a strong feeling, like the impulse to tap one's foot or dance. The exact acoustic properties that give a song swing, however, have long been thought to be undefinable. To investigate swing, a team led by physicist Corentin Nélis delayed the downbeats and synchronized the offbeats in jazz piano solos and asked jazz musicians to compare the intensity of swing in each modified piece with the intensity of swing in the original piece. They found that participants were more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing, suggesting that _____ blank

Which choice most logically completes the text?

- A. synchronized offbeats tend to give a song swing regardless of whether downbeats are delayed.
- B. the acoustic properties that give a song swing are not easy for jazz musicians to manipulate.
- C. jazz songs that feature the piano are more likely to have swing than are jazz songs that do not feature the piano.
- D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

“Mrs. Spring Fragrance” is a 1912 short story by Sui Sin Far. In the story, Mrs. Spring Fragrance, a Chinese immigrant living in Seattle, is traveling in California. In letters to her husband and friend, she demonstrates her concern for what’s happening at her home in Seattle while she is away: _____ blank

Which quotation from Mrs. Spring Fragrance’s letters most effectively illustrates the claim?

- A. “My honorable cousin is preparing for the Fifth Moon Festival, and wishes me to compound for the occasion some American ‘fudge,’ for which delectable sweet, made by my clumsy hands, you have sometimes shown a slight prejudice.”
- B. “Next week I accompany Ah Oi to the beauteous town of San José. There will we be met by the son of the Illustrious Teacher.”
- C. “Forget not to care for the cat, the birds, and the flowers. Do not eat too quickly nor fan too vigorously now that the weather is warming.”
- D. “I am enjoying a most agreeable visit, and American friends, as also our own, strive benevolently for the accomplishment of my pleasure.”

Believing that living in an impractical space can heighten awareness and even improve health, conceptual artists Madeline Gins and Shusaku Arakawa designed an apartment building in Japan to be more fanciful than functional. A kitchen counter is chest-high on one side and knee-high on the other; a ceiling has a door to nowhere. The effect is disorienting but invigorating: after four years there, filmmaker Nobu Yamaoka reported significant health benefits.

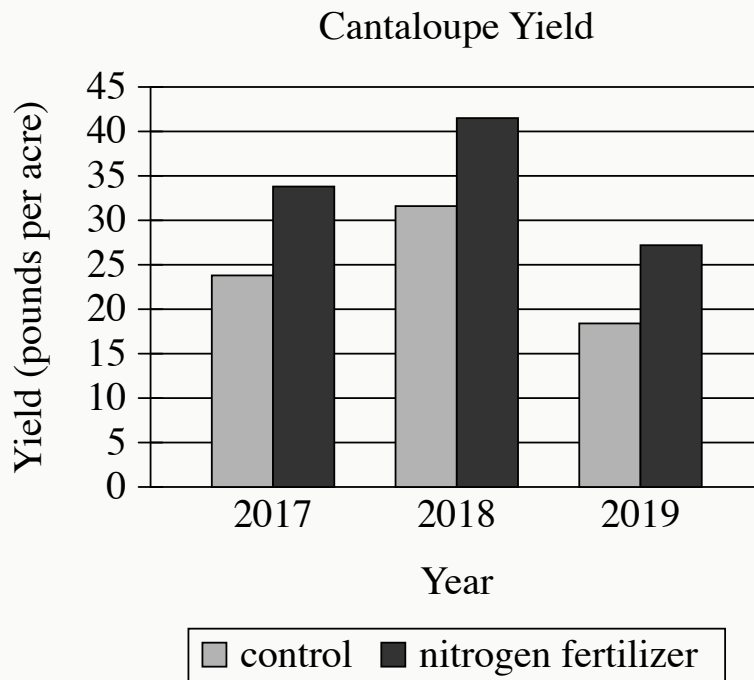
Which choice best states the main idea of the text?

- A. Although inhabiting a home surrounded by fanciful features such as those designed by Gins and Arakawa can be rejuvenating, it is unsustainable.
- B. Designing disorienting spaces like those in the Gins and Arakawa building is the most effective way to create a physically stimulating environment.
- C. As a filmmaker, Yamaoka has long supported the designs of conceptual artists such as Gins and Arakawa.
- D. Although impractical, the design of the apartment building by Gins and Arakawa may improve the well-being of the building's residents.

In Switzerland, the white fuzzy mountain flowers known as edelweiss are widely treated as a symbol of strength and courage. Although edelweiss can thrive in extreme conditions, they aren't notably tougher or harder to reach than other mountain flowers growing in the Swiss Alps. Historian Tobias Scheidegger has shown that the popular view of the flowers originated in the mid-1800s when mountain climbing became popular in Switzerland. Mountain climbers spread the idea that the flowers grew only in steep, icy terrains that were dangerous to climb to. Scheidegger says that these claims were self-interested. He suggests that mountain climbers presented edelweiss in this way in order to _____ blank

Which choice most logically completes the text?

- A. make themselves appear brave and strong for being able to climb to difficult places where edelweiss supposedly grew.
- B. encourage more flower enthusiasts to explore the Swiss Alps.
- C. share their observations about the unusual characteristics of edelweiss with scientists.
- D. prove that edelweiss were more common in the Swiss Alps than in other mountain regions in Europe.



- For each data category, the following bars are shown:
 - control
 - nitrogen fertilizer
- The data for the 3 categories are as follows:
 - 2017:
 - control: 23.8 pounds per acre
 - nitrogen fertilizer: 33.8 pounds per acre
 - 2018:
 - control: 31.6 pounds per acre
 - nitrogen fertilizer: 41.5 pounds per acre
 - 2019:
 - control: 18.4 pounds per acre
 - nitrogen fertilizer: 27.2 pounds per acre

To test the effects of a nitrogen fertilizer on cantaloupe production, researchers grew cantaloupe plants and harvested their fruit over three years. In each year, half the plants were grown using a nitrogen fertilizer, and the other half were grown using a

control fertilizer that contained no nitrogen. The researchers concluded that the nitrogen fertilizer increases cantaloupe yield.

Which choice best describes data in the graph that support the researchers' conclusion?

- A. In every year of the experiment, plants treated with the nitrogen fertilizer had a yield of at least 30 pounds per acre.
- B. In every year of the experiment, plants treated with the nitrogen fertilizer had a greater yield than did plants treated with the control fertilizer.
- C. The 2018 yield for plants treated with the control fertilizer was greater than was the 2019 yield for plants treated with the nitrogen fertilizer.
- D. The yield for plants treated with the nitrogen fertilizer increased from 2017 to 2018.

When Kenyan writer Ngũgĩ wa Thiong’o, who had previously published four novels in English, began writing in his native language, Gĩkũyũ, in the 1970s, several fellow writers and critics cautioned that doing so might make his works inaccessible outside his own community. Some noted that Kiswahili—widely spoken in Kenya and elsewhere in Africa—would be a more practical choice. Rejecting their arguments, Ngũgĩ went on to author dozens of acclaimed works in Gĩkũyũ that have been translated into a total of more than thirty languages.

Which choice best states the main idea of the text?

- A. The reaction to Ngũgĩ’s rejection of English illustrates that some literary experts believe that fame is most easily gained by writing in a widely understood language, such as Gĩkũyũ.
- B. Although Ngũgĩ insisted on publishing his first works in Gĩkũyũ, they have since been translated into many other languages.
- C. Although Ngũgĩ’s decision to write in Gĩkũyũ was met with some skepticism, it didn’t prevent him from achieving literary success.
- D. In the 1970s, Ngũgĩ became convinced that literature ought to be written in authors’ native languages, and he proceeded to publish many works in Gĩkũyũ.